

SHUBHAM PURI

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EDUCATION

New York Institute of Technology, Manhattan, NY,
B.S. in Computer Science

May 2026

Relevant Courses: Data Structures, Design and Algorithms, Software Engineering, Artificial Intelligence, Computer Networks, System Design

SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL

Frameworks/Libraries: React, React Native, Node.js, FastAPI, LangGraph, TensorFlow Lite

Database: PostgreSQL, Supabase, Firebase

Tools & Platforms: Docker, Git, GitHub, Vercel, RESTful APIs, CI/CD, Figma, Jira

PROJECTS

[CS Career Navigator - Personalized Career Path Tool](#) [Demo](#)

Python, FastAPI, Supabase, REST API, Gemini API

- Researched career mentorship gap with 20+ CS students; deployed intelligent assessment tool to 21+ active users, achieving **58% 30-day retention**.
- Defined product roadmap and user stories across 4 Agile sprints; A/B tested recommendation changes, improving relevance by **40%** and reducing time-to-career-clarity to a single session.

[AI-Powered Malware Detection App](#) [Demo](#)

React Native, TensorFlow Lite, Firebase, Python

- Built a full-stack mobile app detecting malware in Android APKs via on-device TFLite neural network, enabling **fully offline** scanning with no user data sent to external servers.
- Implemented a cloud-based REST API model update pipeline, ensuring the detection engine stays current without requiring app store updates, maintaining accuracy against emerging threats.
- Designed an AI agent workflow guiding users through scan results with contextual explanations and step-by-step remediation, reducing friction for non-technical audiences.

[Salon Booking Platform - Production Web App](#) [Demo](#)

TypeScript, Node.js, Supabase, PostgreSQL, REST API, Vercel

- Built and deployed a **production full-stack** booking platform actively used by a real salon business, and conducted user interviews with the owner to define core flows for appointments, service browsing, and admin management.
- Streamlined appointment booking from 7 steps to 3 through iterative client feedback cycles; applied OOP design patterns delivering a scalable, maintainable codebase aligned to real business operations.

[Autonomous Code Review and Testing Agent](#)

Python, LangGraph, tree-sitter, GitHub API, Docker

- Built a LangGraph agentic system that autonomously generates unit test suites, executes them in isolated Docker sandboxes, self-corrects failures, and opens a GitHub pull request.
- Implemented AST-based code understanding via tree-sitter across multiple languages, enabling context-aware test generation covering edge cases beyond simple prompt-based tools.
- Designed a self-correction feedback loop that diagnoses root causes of test failures, patches generated tests, and iterates until a configurable coverage threshold is met.

WORK EXPERIENCE

[Queens Library - Tech Mentor](#)

Queens, NY | May 2025 - Present

- Led STEM workshops in web dev, Arduino, 3D printing, and Lego robotics for **15+** teens; taught Python for hardware programming and introduced AI tools and concepts.
- Designed a 6-week web development curriculum 3 students completed independent projects.

[Outlier - AI Quality Analyst](#)

Remote | Oct 2024 - Feb 2025

- Designed 25+ adversarial and edge-case prompts across code generation, reasoning, and factual accuracy tasks using Agile sprint-based evaluation cycles, contributing to a **20% improvement** in model accuracy.
- Surfaced systemic failure patterns and documented failure, improving model robustness and accelerating the refinement cycle across multiple task categories.

[Mouse Squad - Tech Support](#)

Brooklyn, NY | Sep 2023 - April 2024

- Diagnosed hardware and software issues across a multi-device environment, reducing system downtime by **30%**; managed SQL-logged inventory for 50+ devices, ensuring uninterrupted staff access.

HACKATHONS & COMPETITIONS

[Nepali Hackathon](#)

March 2026 | Team

- Built a mental health web app for stigma-free early detection and peer support; Top 4 of 83 teams

[NYIT Hackathon](#)

Manhattan 2024 | Solo

- Built a real-time facial recognition system using Python and OpenCV; extended into a fully automated attendance pipeline with identity verification.